Prasad Sadanand Mahabare

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TECHNICAL SKILLS

- **Programming: Embedded C, C++, Python, Bash, JavaScript**
- **Microcontrollers: PIC18F4520, STM32F446RE, ESP32-WROOM**
- **Developer Tools: Git, GitHub, VS Code, MPLAB X, STM32CubeIDE, Arduino IDE, Postman**
- **Cloud & DevOps: AWS EC2, Docker, Linux (Ubuntu), Nginx**
- **Automation & Integration: n8n, REST APIs, Webhooks, OAuth 2.0, JSON**
- **Protocols: UART, I2C, SPI, MQTT, HTTP/HTTPS**
- **Databases: PostgreSQL, SQLite**
- **IoT Cloud Platform: Blynk, ThingSpeak, Node-RED**

PROJECTS

ESPForge - OTA Firmware Management Platform | Github

- Engineered **Wi-Fi-enabled ESP32** software for remote **OTA upgrades**, **watchdog-based recovery**, and live device telemetry.
- Integrated **FreeRTOS** task orchestration with **MQTT** messaging for deployment pipelines and remote maintenance.
- Designed a **cloud dashboard** supporting firmware version management, update triggering, and node status tracking.

STM32 FreeRTOS System Health Monitor | Github

- Created a multitasking **STM32 Cortex-M4** application using **FreeRTOS** for deterministic scheduling and runtime task monitoring.
- Combined **UART diagnostics**, **ADC sampling**, heartbeat monitoring, and anomaly detection through concurrent execution flows.
- Configured **GPIO, ADC, UART,** and timer peripherals in **STM32CubeIDE** using priority-driven embedded architecture.

PIC18F4520 UART Command & Peripheral Management System | Github

- Implemented a **UART-controlled** management interface on **PIC18F4520** for command-driven operation of embedded peripherals.
- Architected **interrupt-based serial handling** and parser logic for stable low-latency communication.
- Performed hardware validation through **Proteus simulation**, peripheral testing, and **serial trace debugging**.

PROFESSIONAL EXPERIENCE

Software Developer, Fuge Technologies **May 2024 - Sep 2025**

- Developed and maintained **RESTful APIs** using **Java Spring Boot** to process virtual machine and infrastructure data, supporting **10+ automated workflows**.
- Built backend features using **Java Streams** and optimized data processing logic, reducing response times by **15%** for routine operations.
- Deployed and maintained applications on **AWS EC2** using **Docker**, improving deployment consistency and reducing manual setup effort by **40%**.

Network Engineer, Purdue University **Jan 2023 – Dec 2023**

- Configured and maintained Cisco Catalyst switches (2960–9300 series), firewalls, DNS infrastructure, and load balancing systems supporting enterprise network operations.
- Automated firmware deployment workflows for 800+ network devices using custom scripts, reducing manual upgrade effort and improving deployment consistency.
- Applied security patches, hotfixes, and system updates across Purdue cloud infrastructure to improve operational reliability.

EDUCATION

P.G. Diploma in Embedded Systems Design **Sep 2025 - Jul 2026**

BICARD Institute, Pune

Master of Science in Computer Science **Jan 2022 - Dec 2023**

Purdue University, Fort Wayne, Indiana

Bachelor of Engineering Computer Engineering **Aug 2016 - Aug 2019**

Pune University, Pune, India